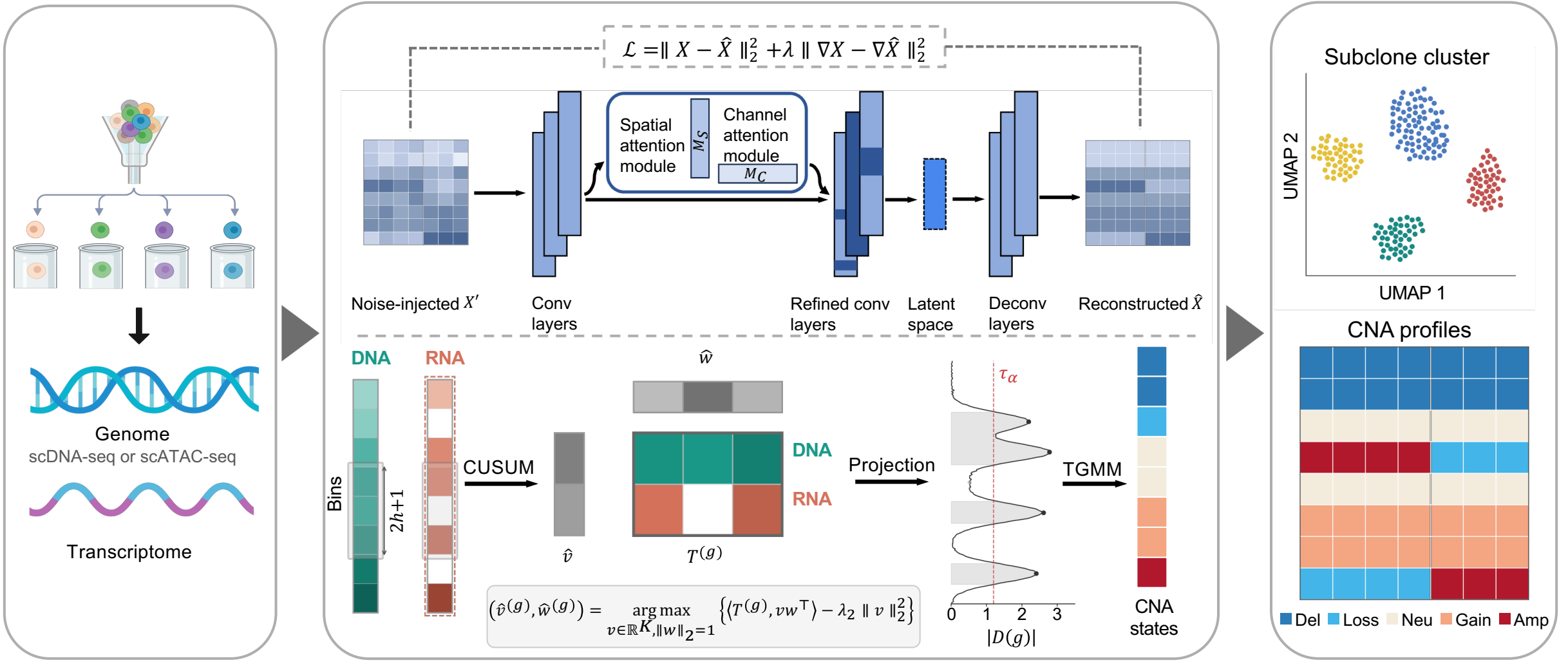

Genome
scDNA-seq or scATAC-seq
Transcriptome
$\mathcal{L} = \| X - \hat{X} \|_2^2 + \lambda \| \nabla X - \nabla \hat{X} \|_2^2$
Spatial attention module
M_S
Channel attention module
M_C
Noise-injected X'
Conv layers
Refined conv layers
Latent space
Deconv layers
Reconstructed $\hat{X}$
DNA
RNA
Bins
$2h+1$
CUSUM
$\hat{v}$
$\hat{w}$
DNA
RNA
$T^{(g)}$
Projection
τ_α
0 1 2 3
$|D(g)|$
TGMM
CNA states
$(\hat{v}^{(g)}, \hat{w}^{(g)}) = \underset{v \in \mathbb{R}^K, \|w\|_2 = 1}{\arg\max} \{ \langle T^{(g)}, vw^\top \rangle - \lambda_2 \| v \|_2^2 \}$
Subclone cluster
UMAP 2
UMAP 1
CNA profiles
Del Loss Neu Gain Amp